



Pharm*Achieve*

HEART FAILURE

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ACEi	Angiotensin-Converting Enzyme inhibitor
AF	Atrial Fibrillation
AKI	Acute Kidney Injury
ARB	Angiotensin Receptor Blocker
ARNI	Angiotensin II Receptor Blocker Neprilysin Inhibitor (Sacubitril/Valsartan)
ASCVD	Atherosclerotic Cardiovascular Disease
BB	Beta Blocker
BMI	Body Mass Index
BNP	Brain Natriuretic Peptide
CABG	Coronary Artery Bypass Surgery
CAD	Coronary Artery Disease
CCB	Calcium Channel Blocker
CHF	Congestive Heart Failure
CKD	Chronic Kidney Disease
CO	Cardiac Output
CRT	Cardiac Resynchronization Therapy
CV	Cardiovascular
CXR	Chest X Ray
DELIVER	Dapagliflozin in Heart Failure with Mildly Reduced or Preserved Ejection Fraction
DPP4i	Dipeptidyl Peptidase-4 Inhibitor
EF	Ejection Fraction
EMPEROR-PRESERVED	Empagliflozin in Heart Failure with a Preserved Ejection Fraction
FINEARTS-HF	Finerenone in Heart Failure with Mildly Reduced or Preserved Ejection Fraction
GDMT	Guideline Directed Medical Therapy
HCN	Hyperpolarization-Activated Cyclic Nucleotide-Gated

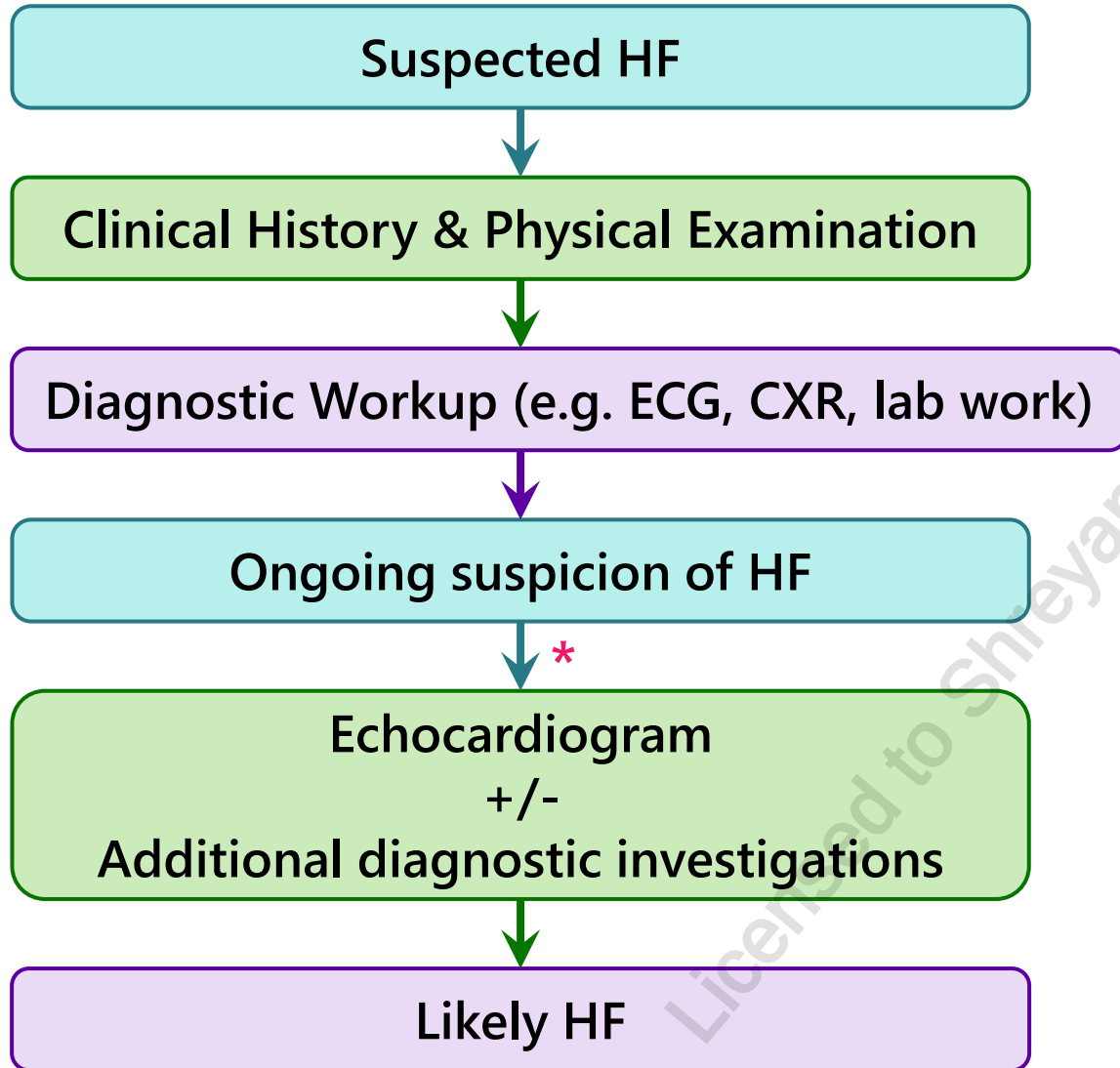
HF	Heart Failure
HFH	Heart Failure Hospitalization
HFmrEF	Heart Failure with Mildly Reduced Ejection Fraction
HFnrEF	Heart Failure with Nonreduced Ejection Fraction
HFpEF	Heart Failure Preserved Ejection Fraction
HFrEF	Heart Failure Reduced Ejection Fraction
HR	Heart Rate
HTN	Hypertension
ICD	Implantable Cardioverter Defibrillator
JVP	Jugular Venous Pressure
LVEF	Left Ventricular Ejection Fraction
N/V/D	Nausea/ Vomiting/ Diarrhea
M	Mortality
MI	Myocardial Infarction
MRA	Mineralocorticoid Receptor Antagonist
MR	Mitral Regurgitation
PARAGLIDE-HF	Prospective comparison of ARNI with ARB Given following stabilization in Decompensated HFpEF
PCI	Percutaneous Coronary Intervention
PM	Pacemaker
RASi	Renin-Angiotensin System Inhibitor
RAAS	Renin Angiotensin Aldosterone System
S	Symptoms
sGC	Soluble Guanylate Cyclase
SGLT2i	Sodium-glucose co-transporter 2 inhibitor
SOB	Shortness of Breath
SR	Sinus Rhythm
SV	Stroke Volume
T2DM	Type 2 Diabetes Mellitus
TNFi	Tumour Necrosis Factor Inhibitors

INTRODUCTION

- **Heart Failure (HF)**: impaired ability of ventricles to fill with (diastole) or eject (systole) blood
- **Decompensated HF**: acute worsening of symptoms caused by volume overload
- **2 types of HF**:
 - ❑ HF with reduced ejection fraction (**HFrEF**) → **LVEF $\leq$ 40%**
 - ❑ HF with non-reduced ejection fraction (**HFmrEF***) → **LVEF $>$ 40%** (*Previously known as HFmrEF and HFpEF)
- **Clinical presentation**:

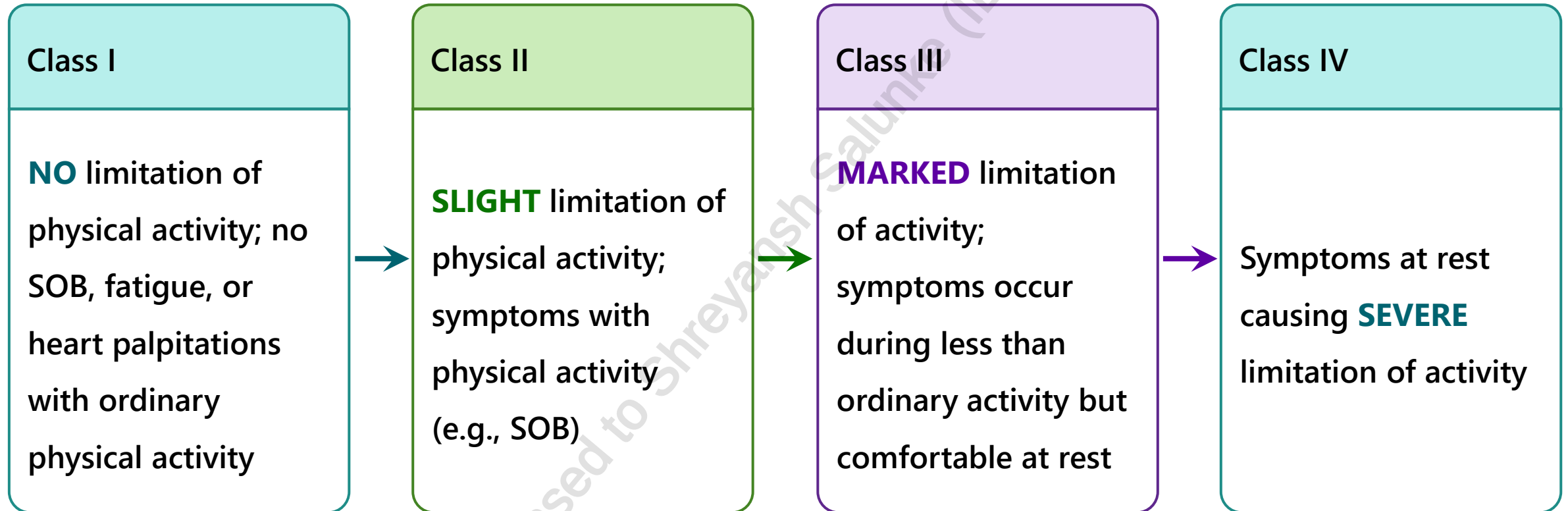
Hypotension	Hypoxia, fatigue, cyanosis, confusion, chest pain
Dyspnea	Sudden nocturnal dyspnea, orthopnea, tachypnea, crackles, cough
Systemic congestion	Jugular venous congestion, hepatomegaly, ascites, nausea, vomiting
Peripheral edema	Weight gain, abdominal distension

DIAGNOSIS



*Natriuretic peptide testing has limited availability in Canada, used to support diagnosis when possible

NYHA CLASSIFICATION OF HEART FAILURE



RISK FACTORS AND CAUSES

- HTN
- Diabetes
- CAD
- Male (HFrEF)
- Smoking
- Increased age
- Obesity
- Dyslipidemia
- Valvular heart disease
- Tachyarrhythmias
- Heavy alcohol use
- Physical inactivity
- Congenital abnormalities
- Infections
- Medications (next slide)



Heart failure is largely preventable. The risk is reduced by 50% by managing HTN alone

DRUG-INDUCED CAUSES AND EXACERBATION OF HF

Mechanism	Pharmacological Classes or Drugs
Fluid retention	• NSAIDs (e.g. celecoxib, ASA $\geq$ 325mg/ day) • Corticosteroids • Gabapentinoids (e.g. pregabalin, gabapentin) • Thiazolidinediones (e.g. pioglitazone) • Black licorice
Reduced cardiac output	• Specific antiarrhythmic drugs (e.g. dronedarone, flecainide, propafenone) • Specific CCBs (e.g. verapamil, diltiazem) • Itraconazole • Overdose of TCAs, carbamazepine
Oncology Drugs	• Anthracyclines (e.g. doxorubicin) • Antimetabolites (e.g. 5-fluorouracil) • Alkylating agents (e.g. cyclophosphamide, ifosfamide) • Taxanes (e.g. paclitaxel) • Interleukin-2
Other	• Clozapine (myocarditis) • Lithium, sympathomimetics (e.g. amphetamines, methylphenidate, cocaine) • Saxagliptin • Hydroxychloroquine, TNFi (e.g. infliximab)

GOALS OF THERAPY

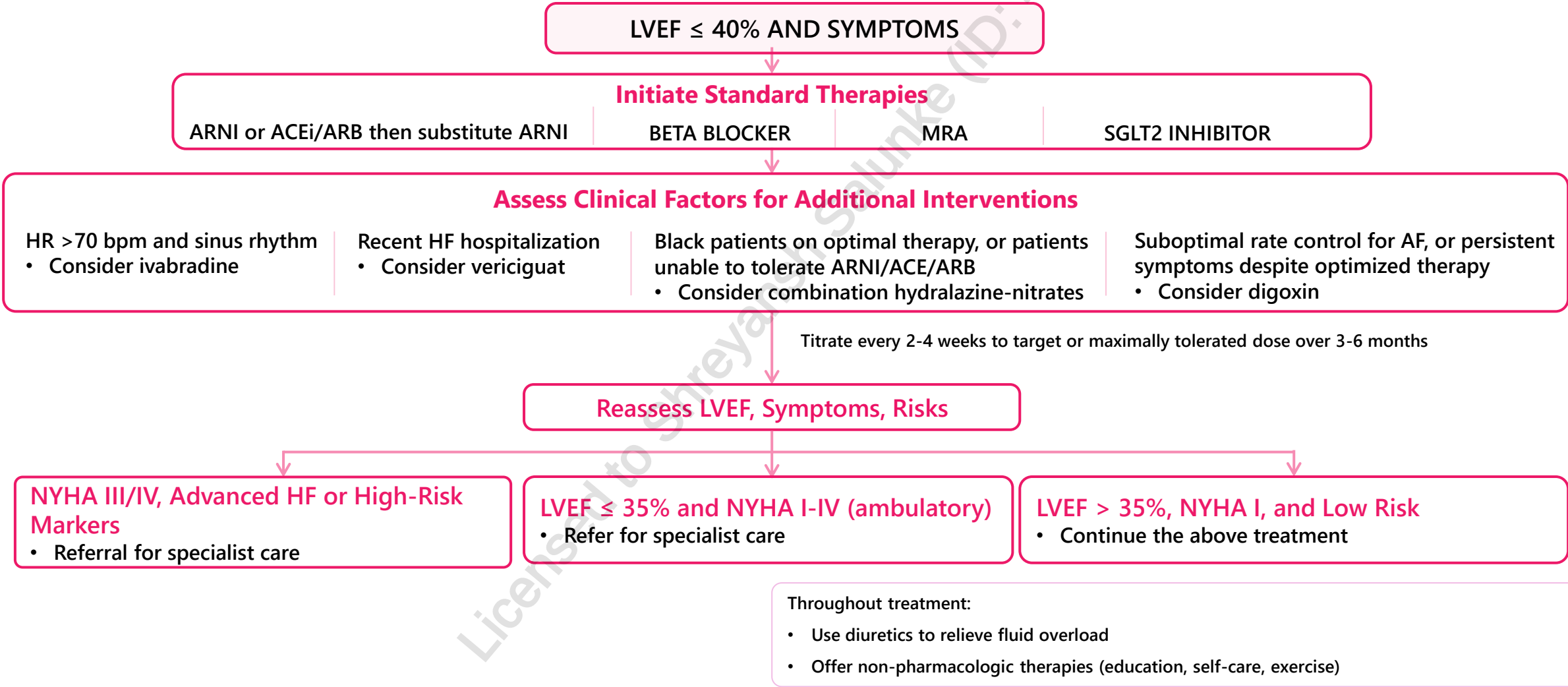
- ✓ Reduce symptoms of heart failure
- ✓ Minimize exacerbations, hospitalizations and morbidity
- ✓ Improve survival, functional capacity (e.g. exercise tolerance), and quality of life
- ✓ Minimize drug-related adverse reactions

NON-PHARMACOLOGICAL MEASURES

- **Salt restrictions:** < 2 - 3 g Na⁺/day or < 1 – 2 g/day if severe heart failure
- **Fluid restrictions:** 2 L/day from all sources
- **Reduce alcohol intake:** recent guidelines have stated that negative health effects are possible with any alcohol consumption
- **Exercise moderately** if stable HF to improve exercise tolerance
- **Monitor weight frequently** (common to gain > 4 kg from edema)
- **Immunizations:** influenza and pneumococcal vaccines to reduce risk of acute exacerbations
- **Treat risk factors:** HTN, diabetes
- **Smoking cessation**
- **PCI or CABG** in symptomatic ischemia
- **ICD** if history of sudden cardiac arrest or ventricular fibrillation

TREATMENT ALGORITHM - HFrEF

Goal of therapy: Reduce symptoms, mortality and hospitalizations



TREATMENT ALGORITHM - HFnrEF

Goal of therapy:
Reduce hospitalizations

LVEF > 40% AND SYMPTOMS

Initiate Standard Therapies

SGLT2i

MRA

Patient Specific Criteria

LVEF $\geq$ 45% & BMI $\geq$ 30 kg/m²

Not at risk of symptomatic
hypotension & SBP $\geq$ 100 mmHg

GLP-1 Receptor Agonists

ARNI (or candesartan,
only if ARNI not
available or tolerable)

Throughout treatment:

- Use diuretics to relieve fluid overload
- Offer non-pharmacologic therapies (e.g. education, exercise)

HF PHARMACOLOGICAL THERAPY – SGLT2I

BENEFITS OF DRUG THERAPY ✓

Drugs	Comments
Dapagliflozin Empagliflozin	Role in therapy: • Improve symptoms and reduce HFH and CV mortality in patients with HFrEF, regardless of T2DM status (<i>DAPA-HR, EMPEROR-Reduced</i>) → best evidence for empagliflozin and dapagliflozin in HFrEF, but canagliflozin may be considered • Reduce risk of HFH in patients with HFmrEF (<i>EMPEROR-Preserved, DELIVER</i>) Pearls: • Not used in patients with HF and type 1 diabetes • Hold SGLT2i on sick days/ dehydrating illness to reduce risk of ketoacidosis • If altered renal function: • Avoid initiating empagliflozin when $\text{eGFR} < 20 \text{ mL/min/1.73 m}^2$ • Avoid initiating dapagliflozin when $\text{eGFR} < 25 \text{ mL/min/1.73 m}^2$ Side effects: • Transient eGFR dip (15 – 25%), generally resolves in 1 – 3 months • Canagliflozin has an increased risk of lower limb amputation • Genital mycotic infections

HF PHARMACOLOGICAL THERAPY – MRA

BENEFITS OF DRUG THERAPY ✓

Drug	Comments
Spironolactone Eplerenone Finerenone	Role in therapy: • Reduce symptoms, CV hospitalizations and mortality in patients with HFrEF (e.g. <i>EMPHASIS-HF</i>) → Only spironolactone and eplerenone are recommended in HFrEF • Reduce risk of HFH in patients with HFmrEF (e.g. <i>FINEARTS-HF</i>) Pearls: • Monitor renal function and electrolytes within 1 week of initiation and any dose change, then every 3 months • Avoid initiating if $K^+ > 5 \text{ mmol/L}$ • If altered renal function: • Avoid initiating spironolactone and eplerenone in $eGFR < 30 \text{ mL/min/1.73 m}^2$ • Avoid initiating finerenone when $eGFR < 25 \text{ mL/min/1.73 m}^2$ Side effects: • MRA can cause hyperkalemia; discontinue MRA if $K^+ \geq 6 \text{ mmol/L}$ • Spironolactone has anti-androgen effects, can cause gynecomastia and impotence • In contrast to finerenone and eplerenone

HF PHARMACOLOGICAL THERAPY – ARNI

BENEFITS OF DRUG THERAPY ✓

Drug	Comments
Valsartan/sacubitril	Role in therapy: - Decreases symptoms, HFH and CV death in patients with HFrEF (<i>PARADIGM-HF</i>) - Patients newly diagnosed with HFrEF should be started on ARNI when possible - Acute decompensated patients with HFrEF should be switched from ACEI or ARB to ARNI once stabilized and before hospital discharge - Reduces HFH in patients with HFmrEF and not at risk of hypotension (weaker evidence per subgroup analysis) Pearls: - Caution with $K^+ > 5.0$ mmol/L, $eGFR < 30$ mL/min/1.73 m², or SBP < 100 mmHg Washout period of 36 h required when switching from ACEi to ARNI - No washout period necessary when switching from ARB Side effects: - Hypotension, hyperkalemia, angioedema

HF PHARMACOLOGICAL THERAPY – ACEI/ ARB

BENEFITS OF DRUG THERAPY ✓

Drugs	Comments
ACEI ARBs	Role in therapy: • ACEi or ARB have no clear role in the early management of acute or worsening HFrEF • But ACEi or ARB should be administered in patients with LVEF $\leq 40\%$ after an acute MI as soon as safely possible • No evidence for ACEi in patients with HFmrEF • Candesartan is preferred in patients with HFmrEF when ARNI can not be used → Reduces HFH Pearls: • Transient decrease in eGFR (of up to 30%) when initiating ACEi or ARB • Consider ARB when ACE is not tolerated (Do NOT combine ACEi + ARB) Side effects: • ACE-induced cough → may switch to ARB or ARNI • Hypotension, hyperkalemia, headaches • Contraindicated in pregnancy

HF PHARMACOLOGICAL THERAPY – DIURETICS

BENEFITS OF DRUG THERAPY ✓

S: Symptoms, M: Mortality, H: Hospitalizations

Drug	Comments
Loop Diuretics Furosemide, bumetanide, Ethacrynic acid	Role in therapy: • Maintain euvolemia and relieve congestion in patients with HFrEF or HFmrEF • Loop diuretics are often recommended, but other diuretics may be added in persistent fluid overload Pearls: • Consider decreasing or discontinuing loop diuretics when starting ARNI to prevent symptomatic hypotension or renal dysfunction from dehydration • Furosemide and bumetanide are sulfa-containing drugs – caution should be used in patients with sulfonamide allergies • Ethacrynic acid can be used instead in sulfonamide-allergic patients Side effects: • Hypokalemia, hypomagnesemia, hyponatremia, hypocalcemia, hypertriglyceridemia, hyperglycemia • Ototoxicity, photosensitivity, dizziness, urinary frequency

HF PHARMACOLOGICAL THERAPY – GLP-1, DUAL GIP/GLP-1

BENEFITS OF DRUG THERAPY ✓

Drugs	Comments
GLP-1: Semaglutide DUAL GIP/GLP-1: Tirzepatide	Role in therapy: • No specific indication for patients with HFrEF but can be considered for comorbidities • Reduces HFH and improves quality of life in patients with LVEF $\geq 45\%$ and BMI ≥ 30 kg/m² • Also appear to improve symptoms Pearls: • Semaglutide (with or without meals) • Do not use tirzepatide with DPP4i or GLP1-receptor agonist Side effects: • N/V/D • Injection site reactions • Rare: acute pancreatitis, gallbladder disease • ↓ gastric emptying

HF PHARMACOLOGICAL THERAPY – BETA BLOCKERS

BENEFITS OF DRUG THERAPY ✓

Drug	Comments
Carvedilol, Bisoprolol	Role in Therapy • Reduces mortality and hospitalization in patients with HFrEF • No indication for use in HFmrEF management but likely used to manage cardiac comorbidities (e.g. AF, CAD) Pearls • Start slow and increase slowly, double the dose every 2 - 4 weeks • Avoid abrupt cessation due to rebound tachycardia • Consider selective agents if concomitant reactive airway disease • Contraindicated in patients with heart block > 1° without pacemaker • If atrioventricular (AV) block is present, consider decreasing other AV node-blocking agents (e.g. amiodarone, digoxin) Side effects: • May mask hypoglycemia symptoms in diabetes (e.g. tachycardia, anxiety, tremors) → monitor for other symptoms (e.g. sweating/ confusion) • Transient fluid retention may occur upon initiation or titration

HF PHARMACOLOGICAL THERAPY – HCN CHANNEL BLOCKER

BENEFITS OF DRUG THERAPY ✓

Drug	Comments
Ivabradine	Role in therapy: - Recommended in patients with symptomatic HFrEF despite GDMT with resting heart rate ≥ 70 bpm in sinus rhythm - Approved in Canada for resting HR of ≥ 77 bpm - No recommendation for use in HFmrEF Pearls: - Used in patients who are intolerant of beta blockers with a resting heart rate ≥ 70 bpm and in sinus rhythm - AVOID in patients with advanced liver disease - Tolerated well in elderly patients Side effects: - Blurred vision and phosphenes (light rings/spots) - onset within 2 months, resolves if stopped - QT prolongation, Bradycardia

HF PHARMACOLOGICAL THERAPY – CARDIAC GLYCOSIDE

BENEFITS OF DRUG THERAPY ✓

Drug	Comments
Digoxin	Role in therapy: • Relieves symptoms and reduces hospitalizations in patients with HFrEF in sinus rhythm who have moderate-severe symptoms despite GDMT • Also used in patients with HFrEF and AF with poor ventricular rate control and/ or persistent symptoms despite beta-blocker optimization • No recommendation for use in HFmrEF Pearls: • Serum digoxin levels < 1.2 ng/mL are associated with lower morbidity, though levels should not be used to guide chronic therapy • Increased risk of digoxin toxicity in patients with reduced renal function, older patients, low body weight, and women • Monitor serum creatine and K⁺ with dose increases in digoxin or during dehydrating illness Side effects: • Atrial and ventricular arrhythmias – risk increases with hypokalemia and renal function decline • Visual disturbances, nausea, vomiting, diarrhea, may cause anorexia

HF PHARMACOLOGICAL THERAPY – SGC STIMULATORS

BENEFITS OF DRUG THERAPY ✓

Drug	Comments
Vericiguat	Role in therapy: - Added to optimal HF therapy in patients with worsening HFrEF symptoms and HFH in the past 6 months (<i>VIRTORIA</i>) → reduces HFH and mortality - No indication in HFmrEF management Pearls - Indicated in high-risk patients that have high rates of hospitalization and mortality - Avoid in patients with SBP < 100 mmHg and eGFR < 15 ml/min/m² Side effects: - hypotension, dyspepsia, nausea, anemia

HF PHARMACOLOGICAL THERAPY – NITRATE/ HYDRALAZINE

BENEFITS OF DRUG THERAPY ✓

Drug	Comments
Isosorbide dinitrate and hydralazine	Role in therapy: • Vasodilation in patients with HFrEF who are unable to tolerate ARNI, ACEi or ARB, due to hyperkalemia, renal dysfunction or other contraindications • Also added to standard GDMT in black patients with HFrEF and advanced symptoms • Decrease symptoms and mortality • No indication in HFmrEF management Pearls: • Avoid with sGC stimulators due to risk of additive hypotension Side effects: • Hypotension, Headache, flushing, weakness • Lupus-like syndrome with hydralazine (less common but serious)

SWITCHING BETWEEN ACEiS, ARBS, AND ARNIs

ACEi



ARNI



36-hour washout period due to an increased risk of angioedema when used together

ACEi



ARB



no washout period

ARB



ARNI



no washout period

Angioedema

- Mechanism: ACEi inhibits bradykinin degradation, leading to accumulation and causing vasodilation and angioedema
- Presentation: Subcutaneous/ submucosal edema (swelling can affect lips, tongue, face and airways), GI pain, diarrhea
- Onset: minutes up to 72 hours
- Switching to ARB: ARBs are relatively safe alternatives (low risk of cross-reactivity, do not directly interfere with bradykinin metabolism)
 - Initiate under close medical supervision and monitor closely for angioedema
 - Due to lingering effects of ACEi, subsequent episodes of angioedema may occur → a washout period *may* be considered only if urgent drug therapy initiation is not warranted
- Switching to ARNI: **Contraindicated**, ARNIs carry similar angioedema risk as ACEIs

ACUTE DECOMPENSATION

Presentation:

- **Sudden worsening of symptoms of HF** → dyspnea, swelling, volume overload, fatigue
- Volume overload assessment:
 - **Clinically**: e.g. pulmonary edema, swelling, elevated JVP and
 - **Objectively**: e.g. current weight vs dry weight to assess how much excess fluid

Treatment:

- **High-dose loop diuretics** (furosemide) are commonly used to remove excess fluid
- Treat underlying cause (if any) of exacerbation (Such as infections, MI, AKI)
- If ...
 - Previously on BB → recommended to CONTINUE (**BB withdrawal can worsen HF**)
 - BB naive → recommend starting once stable (BB can reduce CO)
- **Generally, continue home HF regimen as vitals and clinical context permits**

SPECIAL POPULATIONS - PREGNANCY AND BREASTFEEDING

Pregnancy

- ACEi, ARB, ARNI, MRA are contraindicated in pregnancy
- Ivabradine and SGLT2i should be avoided due to insufficient data
- Drugs of choice include **beta-blockers (except atenolol), digoxin, hydralazine/ISDN, and diuretics**
 - Use diuretics with caution as they can cause a theoretic reduction in placental perfusion
 - Beta-1 selective beta-blockers are preferred; metoprolol has the most evidence

Breastfeeding

- **Beta-blockers** can be used (metoprolol has the most evidence)
- Furosemide, spironolactone, captopril, enalapril, quinapril are also relatively safe
- ACEi preferred over ARB and ARNI due to the limited data available
- Avoid ivabradine as no safety data is available

TAKEAWAY

HFnrEF (LVEF > 40%)	HFrEF (LVEF ≤ 40%)
Pharmacological Management: • SGLT2i • MRA • ARNI (or candesartan) • No at risk of symptomatic hypotension & SBP ≥ 100 mmHg and • GLP – 1 receptor agonist • if LVEF ≥ 45% and BMI ≥ 30 kg/m²	Pharmacological Management: • SGLT2i • MRA • ARNI or ACEI or ARB • Beta blocker Additional Interventions: • Ivabradine • Vericiguat • Hydralazine-nitrates • Digoxin
• Nonpharmacological Management: salt and fluid restrictions, exercise, smoking and alcohol cessation) • Risk factor management: obesity, diabetes, hypertension, drug induced causes	

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
Signs of CHF (e.g. weight, fatigue, dyspnea, peripheral edema)	Improve and stabilize	2-3 days (diuretics improving fluid retention) Weeks-months (effect of ACEi, beta blocker)

Safety Endpoints		
Parameter	Degree of Change	Timeframe
BP, HR, SCr, K+, Mg2+	Within normal limits BP > 90/60mmHg K: 3.5-5mmol/L	Duration of therapy; more frequent in high-risk individuals
Adverse effects of medications (e.g. cough)	Prevented/minimized	Duration of therapy

CASE

HF is a 72-year-old female with 1 week of worsening shortness of breath. HF presents to the ED with SOB on exertion, a dry cough for the past week, and she now needs to sleep in her recliner due to orthopnea. She had flu-like symptoms 2 weeks ago with fever and chills, which have now resolved, but her cough is persistent. She took some Advil Cold and Sinus Nighttime® (ibuprofen, pseudoephedrine hydrochloride and chlorpheniramine maleate) a few times to help her sleep, but has found it ineffective. HF has a past medical history of CAD; she had a CABG 10 years ago, she has had hypertension for the past 20 years and she has dyslipidemia, hypothyroidism and osteoarthritis. HF is currently taking ASA EC 81 mg daily, ramipril 5 mg po daily, metoprolol 25 mg po BID, atorvastatin 20 mg po daily, levothyroxine 75 mcg po QAM, pantoprazole 40 mg po daily, Tylenol Arthritis® 650 mg (acetaminophen) PRN, Advil Cold and Sinus Nighttime® (ibuprofen, pseudoephedrine hydrochloride and chlorpheniramine maleate) and vitamin D3 1000 units po daily. She does not smoke and she drinks alcohol socially. She has an allergy to latex that results in a rash.

Investigations:

ECHO: LVEF 37%	TSH=1.8 (normal)	JVP elevated
BNP = 340 pg/mL (high)	BP=110/58, HR=98, sinus rhythm	Weight = 66 kg (previously 61 kg)
SCr=119	RR=20, sPO2 90% on room air	K+ = 4.0 (3.5-5.0)

1. All of the following reduce CV mortality in HFrEF EXCEPT:

- a) Spironolactone
- b) Carvedilol
- c) Perindopril
- d) Furosemide



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2. Which therapies should HF be started on?

- a) Furosemide and digoxin
- b) Furosemide and spironolactone
- c) Ivabradine and furosemide
- d) Furosemide and valsartan



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SCr=119	RR=20, sPO2 90% on room air	K+ = 4.0 (3.5-5.0)

3. Which of the following parameters should be monitored closely?

- a) Creatinine
- b) Potassium
- c) Volume status
- d) All of the above



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CHANGE LOG

February 2025

- Grammar, spelling and formatting changes made throughout
- Slide 3: Updated Legend
- Slide 9: Updated alcohol advice based on recent guidelines
- Slide 10: Simplified
- Slide 15: New

July 2025

- Rearranged slide order
- Slides 11-15: Significant edits to drug charts
- Slide 20: Added formulation of Omega-3

September 2025

- Slide 15: formatting edits

December 2025/January 2026

- Significant changes made following new guidelines on HFnrEF

April 2026

- Slide 26: Removed fosinopril and ramipril from drugs safe to use in breastfeeding